COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 - Data Structure Algorithms	**SECTION:**	CpE21S2
DATE PERFORMED:	9/16/2026	**DATE SUBMITTED:**	9/17/2026
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ASSIGNMENT 9.1 Binary Search Tree

1. PROCEDURE OUTPUT

Objectives: To create a program in C++ using search trees.

1. To be familiarized with the search tree algorithm.
2. To be able to create a program with search tree algorithm.

Answer the following questions:

1. **What is a binary search tree?**

   A Binary Search Tree (BST) is a tree data structure where each node has at most two children. Values smaller than the parent are placed on the left, while larger values are placed on the right.

2. **Where can binary search tree be used?**

   Binary Search Trees can be used for fast searching, inserting, and deleting data. They are commonly used in databases, dictionaries, and systems that need organized data.

3. **What is a tree traversal**

   Tree traversal is the process of visiting every node in a tree in a specific order. This includes pre-order, in-order, and post-order traversal.

4. **Differentiate a post-order and pre-order traversal give examples.**

   Pre-order traversal visits the root first, followed by the left subtree and then the right subtree. Its order is Root -> Left -> Right. Post-order traversal visits the left subtree first, followed by the right subtree and then the root. Its order is Left -> Right -> Root.

5. **What is a parse tree and its use?**

   A parse tree is a tree structure that shows how a grammar is used to form a given input or expression. The starting symbol is placed at the top as the root, and the symbols below it represent the different parts of the input. These symbols can be terminals, which are the actual values or characters, or non-terminals, which represent groups of symbols. A parse tree also shows the order in which operations should be performed, with the deeper parts of the tree being processed first. It is commonly used in programming and compilers to understand and evaluate expressions correctly.

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3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Purpose of Use: _____

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